

Shortage Lists and Monopsony

Applied Labor Economics - Proposal

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I. Setting

a. context

France maintains regional lists of shortage occupations (*métiers en tension*), in which employers may hire non-EEA¹ nationals without a burdensome labor market test² (Parlement Français, 2025). This policy was first introduced in 2008, and was revised in 2021, 2024³ and 2025.

Under the most recent iteration of the law, non-EEA workers employed in a listed occupation may apply for residency permit extensions or regularization (Parlement Français, 2024; Service Public, 2025). Legal status becomes contingent upon a sector, binding workers to their current role unless they are willing to face significant risk and administrative costs.

b. research question

We presume that firms are aware of non-EEA's workers incentives to be employed in a listed occupation. We posit that firms might exploit this vulnerability to set wages below marginal productivity for this group - we call this the *monopsony effect*. The question we aim to answer is: *are firms undercutting non-EEA individuals in listed occupations, because of the legal advantages involved for these workers?* In other words, does a *monopsony effect* exist ?

Answering this question entails isolating the *monopsony effect* from the other forces that operate in these labor markets. First, taste-based discrimination might depress wages independently of firm bargaining power (*discrimination effect*). Second, labor scarcity in these occupations may push firms to offer wage premiums to attract workers (*shortage premium effect*).

c. related work

The paper we are most closely related to is Signorelli (2024), which analysed the first iteration of the French shortage lists policy. She looks at the outcomes of the policy-induced increase in migrant labor on native outcomes. She finds that wages for migrant workers fell by more than for natives. We believe this last differential is consistent with our hypothesis.

We also build on Naidu, Nyarko, and Wang (2016), which shows that under the UAE's *kafala* system⁴, workers were paid only 51% of their estimated marginal product. A 2011 reform relaxing mobility restrictions raised this share to 72%. Work on German administrative data (Amior and Stuhler, 2024) shows that immigration strengthens firm monopsony power at the bottom of the pay distribution. These papers validate our core intuition, *i.e.* that wage markdowns can be driven by a collapse of workers' outside options, and that this collapse can be institutionally driven.

De facto, we will extend Signorelli (2024)'s analysis using more recent data (2019-2025). We differ in our conceptual framework, as we try to bring in elements from the migration-monopsony literature into this policy evaluation. Furthermore, we are alone in explicitly disentangling taste-based discrimination from monopsony effects. To do so is essential. Indeed, work such as Glover, Pallais, and Pariente (2017) shows how bias toward minority workers generates observable wage penalties independently from productivity or bargaining power.

¹European Economic Area. Commercial agreement extending the EU common market to Iceland, Norway, and Liechtenstein.

²A regulatory requirement. Employers must demonstrate they cannot find suitable domestic workers before they are allowed to hire foreign workers.

³Minor change, concerning the agricultural world.

⁴A labor sponsorship mechanism that binds migrant workers' legal status and employment rights to a specific employer. It prevents them from changing jobs or leaving the country without their sponsor's consent.

II. Analysis

a. conceptual framework

Wages result from Nash bargaining, and are a function of the firm’s bargaining power. As the firm’s power increases, the negotiated wage converges toward the worker’s outside option. Given that natives and EEA citizens can freely switch employers or occupations, their outside option approximates the competitive market wage. On the other hand, non-EEA citizens in listed occupations work under permits that are forfeited upon leaving their job, thus their outside option is undocumented work⁵. This framework predicts a wage differential between EEA/natives and non-EEA. It is testable.

b. empirical strategy

We wish to retrieve the causal effect on wages of being a non-EEA national in a listed occupation, net of taste-based discrimination and shortage premium effects. To do so, we plan to run a triple-differences regression on a culturally-matched sample. Indeed, the analysis should be limited to non-EEA and EEA nationalities that differ primarily on permit constraints, not on culture or religion⁶. This enables us to attribute the estimates to monopsony rather than discrimination.

The regression will exploit variation in listing status across occupations, regions, and time, comparing wage changes for non-EEA workers relative to natives around listing events. Formally:

$$\ln(w_{ijrt}) = \beta_0 + \beta_1 \text{NonEEA}_i + \beta_2 \text{EEA}_i + \beta_3 \text{Listed}_{jrt} + \beta_4 (\text{NonEEA}_i \times \text{Listed}_{jrt}) \\ + \beta_5 (\text{EEA}_i \times \text{Listed}_{jrt}) + \mathbf{X}'_{ijrt} \gamma + \delta_{jrt} + \varepsilon_{ijrt}$$

where w_{ijrt} is the wage of worker i in occupation j , region r , at time t . NonEEA_i and EEA_i are indicator variables for nationality group, with natives as the omitted category. Listed_{jrt} indicates whether the occupation is on the list. $\mathbf{X}_{ijrt}$ is a vector of individual controls including age, gender, experience, tenure, and full-time status. δ_{jrt} denotes fixed effects, which will be different across specifications⁷.

The coefficient of interest is $\beta_4 - \beta_5$: the differential effect of listing on wages for non-EEA workers relative to EEA workers (both compared to natives). Under our monopsony hypothesis, we expect $\beta_4 - \beta_5 < 0$.

Identification rests on four key assumptions. First, absent listing, wage trends for non-EEA workers in eventually-listed occupations would track those of natives and EEA workers. Second, listing effects are occupation-specific and do not spillover to unlisted occupations. Third, the policy does not alter the composition of non-EEA workers entering listed occupations, so under positive selection, any remaining bias moves downward. Fourth, listing is exogenous. These assumptions should be tested, where possible.

c. data

We will focus on the 2021 reform⁸ and on the minor 2024 reform. The samples will cover 2019-2025, comprise all French regions, and all occupations.

The primary data source is ForCE (*Formation, Chômage et Emploi*). We plan on using the last or the second-to-last vintage, depending on quality. Worker origin and socio-demographic characteristics are drawn from the *Fichier Historique* table, where the statistical unit is the unemployment spell, while wages and job characteristics are drawn from the *Mouvement de Main-d’Œuvre* table, where the statistical unit is the contract. For detailed variable information, please consult the Appendix.

Workers may hold multiple contracts over the period, thus the final dataset will be structured at the contract level. Worker characteristics will be merged from FH using the ForCE key. This implies that workers with multiple contracts will appear several times - another specification, weighted by worker, might be used to account for this.

⁵or the cost of leaving the country, or the risk-weighted administrative cost of getting a new permit.

⁶Please consult the Appendix for the exact details.

⁷occupation $\times$ region $\times$ year is the best way to absorb the *shortage premium effect*. It is however collinear with the Listed indicator, thus the analysis should be repeated with other FE sets.

⁸We will try to study the 2025 one as well, but the most recent ForCE vintage is S12025.

References

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Appendix

a. culturally matched samples

Table 1: *Fichier Historique* NATION codes and matched groupings, with bias assessment.

Non-EEA		EEA		bias?
code	comments	code	comments	
24	YUG (34,000)	27 + 96	HRV + SVN (3,873)	Partial
24	YUG (34,000)	97 + 98	BGR + ROU (164,859)	Partial
26	TUR (111,043)	19 + 97 + 98 + 2	GRC + BGR + ROU + CYP (171,348)	Religious
29	Autres européens (133,339)	91 to 95	Baltics + Central EU (61,839)	None

NB: *YUG* comprises of non-EU former Yugoslavia (Montenegro, Serbia, Bosnia and Herzegovina, Kosovo, North Macedonia). With the exception of Kosovo, these countries have virtually the same ethno-linguistic background. Some bias might be introduced because some Kosovar Albanians and Bosniaks may practice Islam. *YUG* is also comparable with Balkan Orthodox countries formerly under Ottoman rule, *ROU* and *BGR*.

b. variables to extract

Ficher Historique

variable	description
force_id	Unique identifier.
SEXE	Sex of the job seeker.
REGION	Region code of residence.
NATION	Nationality code.
DATNAIS	Date of birth.
TEMPS	Work time for the job sought.
CONTRAT	Type of contract sought by the job seeker.
ROME	ROME code for the job sought.
EXPER	Experience in the sought ROME.
EXPEUNIT	Unit of experience duration.
DIPLOME	Diploma obtained by the job seeker at registration.
QUALIF	Job seeker's qualification level.
NIVFOR	Education level attained by the job seeker.
NDEM	Demand number (numbered from most recent to oldest).
DATINS	Demand opening date.
DATANN	Demand cancellation date.

Mouvement de Main d'Oeuvre

variable	description
force_id	Unique identifier.
DebutCTT	Contract start date.
FinCTT	Contract end date.
H.Etab_SQN	Establishment identifier.
L.contrat_sqn	Contract identifier.
ModeExercice	Employment time status.
salaire_base	Base salary.
salaire_base_mois_complet	Base salary for a full month of work.